

TECHNICAL RESEARCH REPORT

Integrated Theoretical Framework for Byproduct Reintegration in Closed-Loop Pyrolysis Systems: A Literature-Based Theoretical Synthesis

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September 2026

Abstract

This technical report presents a literature-based theoretical synthesis for establishing closed-loop pyrolysis systems. The research investigates the thermochemical pathways of macromolecular decomposition for biomass and synthetic polymers.

A quantitative mass and heat balance analysis, reproduced from a theoretical balance for one oil sludge, reports that a recirculating solid heat carrier process can achieve a whole-process heat-balance efficiency of 73.6%.

Furthermore, the study explores the catalytic role of byproduct char in vapor upgrading and the use of non-condensable gas (NCG) recirculation to modify the reactive atmosphere.

This synthesis provides a theoretical basis for addressing energy security and waste management challenges and discusses the monitoring and control requirements needed to manage the non-linear feedback loops inherent in circular chemical systems.

Keywords: Pyrolysis, closed-loop systems, biochar, syngas, thermochemical conversion, circular biorefinery, heat integration, process control.

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1 Introduction

The pursuit of sustainable thermochemical conversion technologies requires moving toward sophisticated biorefinery architectures where pyrolysis byproducts serve as active process utilities.

Pyrolysis, defined traditionally as the anaerobic thermal decomposition of carbonaceous materials, produces three distinct phases: solid carbonaceous char, condensable liquid bio-oil, and non-condensable gas (NCG).

While traditional open-loop systems prioritize the maximization of a single product, closed-loop theoretical frameworks emphasize the internal reintegration of these byproducts to achieve autothermal stability, enhance product quality, and minimize external energy inputs [3].

This report provides a theoretical analysis of the chemical mechanisms, thermodynamic balances, and system-level integrations required to establish these cycles.

2 Fundamentals of Pyrolysis Mechanisms and Thermochemical Pathways

To understand byproduct reintegration, the fundamental chemical pathways governing macromolecular decomposition must be established [7].

2.1 Macromolecular Decomposition

Lignocellulosic biomass is a heterogeneous composite of cellulose, hemicellulose, and lignin, each exhibiting a unique thermal signature.

Table 1 outlines the decomposition ranges and primary mechanisms for these components alongside synthetic plastics [2, 7].

2.1.1 Hemicellulose

Hemicellulose is the most thermally labile pseudo-component, degrading between approximately 200°C and 350°C [7].

Cleavage of side chains yields acetic acid, carbon dioxide, and light oxygenates.

2.1.2 Cellulose

Cellulose resists thermal breakdown until approximately 300°C to 400°C [7].

The mechanism is dominated by the competition between glycosidic bond cleavage and dehydration.

Competing pathways of mid-chain fragmentation yield low-molecular-weight products (LMWPs).

2.1.3 Lignin

Lignin decomposition occurs over a broad temperature range from approximately 250°C up to 900°C [7].

Cleavage of aryl-ether linkages generates reactive phenolic radicals that can escape as bio-oil vapors or repolymerize into secondary char.

2.2 Representative Decomposition Ranges

Table 1: Representative Decomposition Ranges and Dominant Products

Pseudo-component	Decomposition range	Primary mechanism	Dominant products
Hemicellulose	200–350°C	Side-chain cleavage / Fragmentation	CO_2 , acids, furans
Cellulose	300–400°C	Glycosidic cleavage / Dehydration	Levoglucosan, LMWPs
Lignin	250–900°C	Aryl-ether cleavage / Radical coupling	Phenolics, char, light gases
Plastics (e.g., PE, PP)	400–550°C	Thermal cracking (chain scission)	Aliphatics, waxes

2.3 Simplified Lumped Kinetic Mechanism

The transformation of biomass into final products is mathematically represented using lumped kinetic models [7].

A standard primary decomposition model includes parallel first-order reactions for the feedstock converting to gas, vapor, and char, followed by secondary cracking.

$$\frac{dC_{\text{feed}}}{dt} = -(k_{\text{char}} + k_{\text{gas}} + k_{\text{vap}}) C_{\text{feed}} \quad (1)$$

$$\frac{dC_{\text{vap}}}{dt} = k_{\text{vap}} C_{\text{feed}} - (k_{\text{crack,gas}} + k_{\text{sec,char}}) C_{\text{vap}} \quad (2)$$

$$\frac{dC_{\text{char}}}{dt} = k_{\text{char}} C_{\text{feed}} + k_{\text{sec,char}} C_{\text{vap}} \quad (3)$$

$$\frac{dC_{\text{gas}}}{dt} = k_{\text{gas}} C_{\text{feed}} + k_{\text{crack,gas}} C_{\text{vap}} \quad (4)$$

Here, C_i are mass concentrations, k_i are first-order rate constants, A_i is the pre-exponential factor, $E_{a,i}$ is the activation energy, R is the gas constant, and T is the absolute temperature.

The rate constants are represented by the Arrhenius relationship:

$$k_i = A_i \exp\left(-\frac{E_{a,i}}{RT}\right) \quad (5)$$

Because k_i is a rate constant dependent on temperature, limiting secondary vapor reactions requires strict control of vapor residence time and temperature rather than altering the kinetic constants [6].

2.4 Synthetic Polymer Co-Pyrolysis

Synthetic polymers such as polyethylene (PE) and polypropylene (PP) undergo thermal cracking through chain scission at approximately 400°C to 550°C, producing aliphatic compounds and waxes [2].

Because PE and PP leave little solid residue, a char-based heat carrier loop cannot function on pure plastic feedstock.

Therefore, polymers are often co-pyrolyzed with biomass, for example, the 75% pine / 25% refuse-derived fuel (RDF) blend of Chavando et al. [3], to utilize the resulting biochar.

Alternatively, an external heat source or solid heat carrier may be required, as in fluidized-bed and two-reactor circulating-bed designs [2].

Additionally, sacrificial tire char has been utilized as a catalyst for steam reforming of waste plastics [5].

3 Thermodynamics of Closed-Loop Pyrolysis Systems

A robust closed-loop system must achieve an energy balance where internal product streams fulfill the endothermic enthalpy requirements of the primary reactions [3].

3.1 Enthalpy of Reaction and Specific Heat Capacity

The total specific heat requirement, q_{total} , in J/kg, accounts for sensible heat, latent heat of moisture vaporization, and the net enthalpy of chemical transformations.

$$q_{\text{total}} = \int_{T_{\text{amb}}}^{T_{\text{pyro}}} C_p(T) dT + x_{\text{H}_2\text{O}} \Delta H_{\text{vap}} + \Delta H_{\text{rxn}} \quad (6)$$

where $x_{\text{H}_2\text{O}}$ is the moisture mass fraction and ΔH_{vap} is the latent heat of vaporization in J/kg.

Primary pyrolysis is generally endothermic; however, secondary charring and vapor-phase cracking may be exothermic [7].

For context, by the present author’s calculation from Zhang et al. [8], approximately 93% of the rotary-reactor heat demand in their theoretical sludge balance (2.33 of 2.49 GJ) goes to heating, vaporizing, and superheating the feedstock’s water.

This highlights why the moisture term is critical.

3.2 Material and Heat Balances for Solid Heat Carriers

To demonstrate thermal integration, a theoretical balance for one oil sludge is used as an example [8].

The study deduced the balance theoretically from laboratory analysis.

This differs from low-ash lignocellulosic biomass, which would rely on recirculating biochar or an inert sand medium.

In the modeled process, hot ash at 800°C is mixed with the sludge in a rotary reactor at 520°C.

The resulting semi-coke (solid char) and retorting gas (pyrolysis gas) are burned, with liquefied petroleum gas (LPG) as supplementary fuel, in a fluidized-bed combustor that reheats the ash.

3.3 Heat and Mass Balance Parameters

Table 2: Heat and Mass Balance Parameters per 1000 kg of Received-Basis Oil Sludge

Parameter	Value	Function
Recirculating ash	5940 kg	Sensible heat carrier (800°C ash cooling toward the 520°C reactor temperature)
Supplementary fuel (LPG)	25 kg (1.25 GJ)	Steady-state fuel for the fluidized-bed combustor, needed because semi-coke yield is low
Input heat	13.65 GJ	Feed chemical heat, preheated air, LPG
Output heat	10.04 GJ	Pyrolysis oil, water, ash and flue-gas heat (oil is about 6.45 GJ)
Heat loss	3.61 GJ	Mainly settling, condensation and combustor wall losses
Thermal efficiency	73.6%	Output heat divided by input heat

Note: Semi-coke refers to the solid char residue from pyrolysis. Retorting gas refers to non-condensable pyrolysis gas. LPG refers to liquefied petroleum gas.

The source’s Table 12 prints an output total of 10.44 GJ, while its abstract and text give 10.04 GJ. This report uses 10.04 GJ.

3.4 Thermal Efficiency

The 73.6% figure is a whole-process heat-balance efficiency: total heat leaving in the output streams divided by total heat entering.

It includes sensible heat carried by condensed water and flue gas, so it is not the fraction of feed energy converted to fuel [8].

By the author’s calculation from the balance in Zhang et al. [8], the pyrolysis oil’s chemical heat (6.44 GJ) is approximately 66% of the chemical heat input (8.51 GJ sludge plus 1.25 GJ LPG).

4 Byproduct Reintegration as Reactive Agents

Byproducts can be reintegrated as chemically active agents. Depending on the target product, such as bio-oil or upgraded gas, different operational modes must be applied.

4.1 NCG Recirculation (Oil-Oriented Mode)

Recycling non-condensable gas (NCG) as the carrier or fluidizing gas alters the reactive atmosphere.

While full in situ hydrodeoxygenation (HDO) requires high hydrogen pressures and distinct catalysts, atmospheric recirculation of CO- and CO₂-rich NCG may suppress undesirable secondary decarbonylation and decarboxylation reactions.

Suppressing these pathways leaves oxygen in the oil, which likely raises overall carbon retention and liquid yield.

The hypothesis that this mechanism stabilizes the primary vapors and reduces downstream upgrading needs remains an area for future experimental validation.

4.2 Char-Catalyzed Vapor Upgrading (Gas-Oriented Mode)

Carbonaceous chars can catalyze tar removal.

For instance, sacrificial tire char has been used in a second-stage steam-reforming step to reduce tar from biomass volatiles pyrolyzed at 500°C [1].

While this confirms the efficacy of tire char with steam additions, the direct use of in-loop biochar to upgrade vapors without steam remains a theoretical target that requires further empirical validation.

5 Systems Analysis: Integrated Looping Models

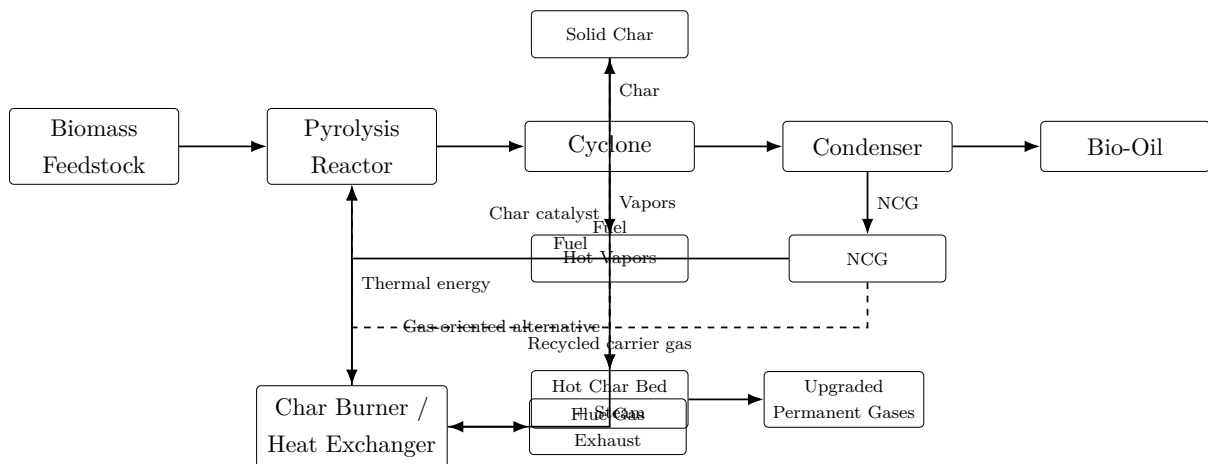
Achieving the required heat transfer necessitates advanced reactor configurations.

Figure 5.1 illustrates the conceptual process flow for byproduct feedback.

5.1 Integrated Process Flow Diagram

The process diagram distinguishes the oil-oriented mode, represented by solid arrows, from the gas-oriented alternative, represented by dashed arrows.

Non-condensable gas is split between recycled carrier gas and burner fuel.



Solid arrows represent the oil-oriented mode, while dashed arrows represent the gas-oriented alternative.

NCG is split between recycled carrier gas and burner fuel. The diagram is a conceptual schematic by the author.

Solid arrows represent the oil-oriented mode, while dashed arrows represent the gas-oriented alternative.

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Figure 1: Conceptual process flow diagram for an integrated closed-loop pyrolysis system. Solid arrows represent the oil-oriented mode, while dashed arrows represent the gas-oriented alternative.

NCG is split between recycled carrier gas and burner fuel. The diagram is a conceptual schematic by the author.

5.2 Comparison of Heating Strategies

Table 3 summarizes the comparison of primary heating strategies utilized in these integrated models.

Table 3: Comparison of Heating Strategies

Strategy	Heat Source	Expected Oil Yield	Dilution / Complexity
External Burner / Solid Heat Carrier	Sensible heat from recirculating ash [8] or sand heated by byproduct combustion.	High (strict anaerobic environment).	High mechanical complexity (dual vessels). No N ₂ dilution in primary reactor.
Autothermal	Partial in situ combustion with air as the oxidant.	Moderate (some product burns for heat).	Low complexity. NCG is highly diluted by N ₂ and CO ₂ .
Chemical Looping	Metal oxide oxygen carrier provides lattice oxygen [9].	Variable (dependent on oxide reactivity).	High complexity. Nitrogen-free NCG without direct air contact.

Note: Expected oil yields are qualitative.

6 Process Control and Optimization Logic

Operating a closed-loop system requires dynamic control over residence time and temperature to manage feedback loops.

6.1 Residence Time vs. Product Targeting

For maximizing bio-oil production, vapor residence time is typically kept to a few seconds at temperatures between 450°C and 550°C to prevent secondary cracking [7].

6.2 Reactor Energy Balance and Control Considerations

Assuming a lumped, well-mixed system, the reactor energy balance dictates the temperature rate of change:

$$m_{\text{bed}} C_{p,\text{bed}} \frac{dT}{dt} = \dot{Q}_{\text{carrier}} + \dot{Q}_{\text{burner}} - \dot{Q}_{\text{feed}} - \dot{Q}_{\text{rxn}} - \dot{Q}_{\text{loss}} - \dot{Q}_{\text{vap}} \quad (7)$$

where m_{bed} is the bed mass, $C_{p,\text{bed}}$ is the specific heat capacity of the bed, $\dot{Q}_{\text{carrier}}$ and $\dot{Q}_{\text{burner}}$ are heat inputs, $\dot{Q}_{\text{loss}}$ is heat lost to the environment, and $\dot{Q}_{\text{vap}}$ is the enthalpy carried away by exiting vapors.

The term $\dot{Q}_{\text{feed}}$ represents the sensible heating of the feed and moisture evaporation, while $\dot{Q}_{\text{rxn}}$ represents the heat of reaction.

The extreme non-linearity of the system arises primarily from the temperature-dependent Arrhenius terms embedded within $\dot{Q}_{\text{rxn}}$.

This dynamic behavior could be managed with advanced control algorithms, such as proportional-integral-derivative (PID) controllers with feedforward compensation.

Monitoring can be implemented with modular monitoring and control electronics [4].

7 Discussion: Theoretical Limitations and Practical Bottlenecks

7.1 Tar Bottlenecks

Although recirculating char mitigates some tar formation, heavy polyaromatics may persist.

In practical applications, tar fouling can remain a bottleneck requiring stringent secondary removal techniques [1].

7.2 Safety and Atmospheric Risks

Recycling CO and H₂ introduces severe hazards, including CO toxicity and the risk of catastrophic ignition if oxygen ingresses into the flammable NCG loop.

Operating outside the flammable range, for example by keeping oxygen below the limiting oxygen concentration, is essential.

7.3 Economic Viability

Evaluating the economic sustainability of closed-loop systems requires dedicated techno-economic assessments.

For example, Chavando et al. [3] incorporate a techno-economic analysis based on their experimental energy balances.

Further optimization at scale remains a critical area for future research.

8 Conclusions and Actionable Insights

Theoretical byproduct reintegration is a promising strategy for enhancing pyrolysis efficiency.

1. Energy Autonomy:

Energy autonomy can be approached by internalizing byproduct energy.

In the 75% pine / 25% RDF blend experiments, byproduct energy yielded 1.750 kW/kg against a primary reactor demand of 1.220 kW/kg, creating a roughly 43% surplus [3].

However, for pine alone, the surplus was only approximately 12% (1.453 kW/kg yield versus 1.298 kW/kg demand), suggesting autonomy is marginal without RDF and could be erased by large-scale heat losses.

This also contrasts with the theoretical oil sludge balance, which still required a 25 kg LPG supplement [8].

2. Product Selectivity:

Char-catalyzed cracking shifts secondary reactions toward gas at the expense of bio-oil.

The hypothesized benefits of NCG-mediated atmospheres on carbon retention and bio-oil stabilization require future validation as an alternative to ex situ upgrading.

3. Future Research:

Validation of direct biochar-catalyzed upgrading without steam, dynamic simulation of closed-loop feedback systems, and detailed techno-economic assessments remain critical next steps.

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